

EduGenie: Google Gemini Powered Learning Assistant

1. Brainstorming & Ideation

This document records the ideation and problem definition for EduGenie. The supplied project description presents EduGenie as a lightweight AI-powered educational assistant intended to simplify learning for students of different academic levels.

1.1 Project Idea

EduGenie is conceived as a learning companion that uses generative AI to make educational interaction simpler, faster, and more accessible.

The project focuses on five core learner needs documented in the source: asking questions, simplifying complex concepts, generating quizzes, receiving personalized learning recommendations, and summarizing large educational passages.

1.2 Problem Statement

Learners may find complex educational material difficult to understand or may need quick explanations and revision material. EduGenie addresses this by combining question answering, explanations, quizzes, summaries, and learning recommendations in one lightweight interface.

The documented design also aims to reduce hardware barriers by using a simple HTML/CSS frontend, FastAPI backend, lightweight local AI for explanations, and cloud-based Gemini capabilities for other tasks.

1.3 Target Users

Students and self-learners across academic levels.

Beginners who need simple explanations of complex concepts.

Learners who want quick revision through summaries and quizzes.

Learners exploring a subject and looking for a structured beginner-to-advanced learning path.

1.4 Documented Use-Case Scenarios

Ocean and river question: a student asks which is the largest ocean.

Pythagoras theorem quiz: a student uses Generate Quiz to assess understanding.

SQL learning path: a learner requests a structured beginner-to-advanced path with timelines and resources.

1.5 Ideated Core Features

Question and answer interaction.

Concept explanation in simplified language.

Three-question MCQ quiz generation with four options per question.

Long-passage summarization.

Personalized learning recommendations from beginner to advanced level.

1.6 AI Strategy

The source documents Gemini 1.5 Pro via API for Q&A, summarization, quiz generation, and learning paths.

The source documents LaMini-Flan-T5-783M as the local model for concept explanation because it is instruction-tuned, lightweight, and CPU-compatible.

Conclusion

The ideation phase establishes EduGenie as a modular educational assistant rather than a single-purpose chatbot. The documented scenarios directly map learner needs to the five core modules.

Source basis: EduGenie project document supplied with this request and the repository structure shown in the supplied reference image. Where the source did not specify an implementation detail, this documentation preserves the documented scope rather than inventing project facts.